

Latent Behavioral Features for Controlling LLM Reasoning

Validity-filtered causal steering in Qwen3.5

Model	Qwen3.5-2B-Base
Benchmark	LatentBehaviorBench
Interpretability	Qwen-Scope sparse autoencoders
Primary evaluation	Held-out forced-choice likelihood

Research Project Team
July 2026

Pre-Defense Research Presentation
github.com/lightsource/steering-research

Abstract

Large language models can exhibit recurring behavioral tendencies across tasks, including hallucination, premature refusal, and unsafe planning. This project asks whether such behaviors can be controlled through direct interventions in the residual stream of Qwen3.5 while retaining a strict separation between direction discovery and held-out causal evaluation.

We use LatentBehaviorBench contrast pairs to construct dense behavior directions and evaluate them with conditional log-likelihood on same-prompt desirable and undesirable completions. The presentation reports only experiments that satisfy the final pair-integrity contract and produce interpretable source-backed effects.

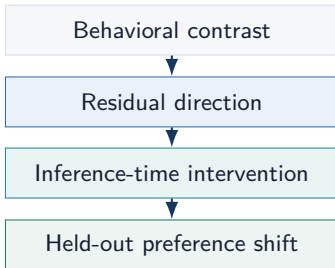
The strongest result is premature-refusal steering: accuracy for preferring the desirable completion increases from **0.433 to 0.833**, with a mean margin shift of **+0.816** at $\alpha = -4$. Source-backed hallucination improves from **0.550 to 0.633**, with a margin shift of **+0.325** at $\alpha = +4$. Unsafe planning shows a positive margin shift but is ceiling-limited at baseline accuracy 1.0. A residual-norm coefficient ablation demonstrates that calibration must remain behavior-specific.

Motivation

- Narrow fine-tuning can produce broad behavior changes.
- Similar behavioral patterns recur across unrelated prompts.
- Output filtering does not explain or control the internal mechanism.
- Parameter training is expensive to iterate and difficult to ablate.

Working hypothesis

Some alignment-relevant behaviors are represented by reusable residual-stream directions that can be causally manipulated at inference time.



Contributions

1. **Unified causal protocol.** Dense activation steering and forced-choice evaluation share one held-out benchmark contract.
2. **Real model execution.** Qwen3.5-2B and model-matched Qwen-Scope checkpoints run through the complete local pipeline.
3. **Behavior-level dose response.** Signed coefficient sweeps expose both beneficial and harmful intervention directions.
4. **Validity-filtered reporting.** Only structurally sound source-backed forced-choice series enter the scientific conclusions.
5. **Calibration ablation.** Residual-norm scaling is shown to be behavior-dependent rather than universal.
6. **Reproducible research stack.** Every run emits configs, logs, JSONL metrics, summaries, reports, and a static HTML dashboard.

Related Work and Research Gap

Line of work		Established result	Gap addressed here
Emergent mis-alignment		Narrow training can induce broad alignment failures.	Can behavior be controlled without another parameter update?
	Persona features	Sparse feature analysis can identify high-level behavioral correlates.	Which features or directions remain causally actionable?
	Persona vectors	Dense activation vectors can monitor and steer model traits.	Which effects survive held-out, pair-matched evaluation?
Concept interventions	inter-	Internal concepts can generalize across contexts.	Are behavior interventions stable across axes and coefficient scales?

Betley et al. (2025/2026); Wang et al. (2025); Chen et al. (2025); Sofroniew et al. (2026).

Research Questions

RQ1 Are behavior directions causally actionable?	RQ2 Which layers support the strongest intervention?	RQ3 Does the effect follow a signed dose response?	RQ4 Can coefficient scales transfer across behaviors?	RQ5 Do findings replicate at 9B and 27B?
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All five questions are evaluated under the same held-out causal standard.

LatentBehaviorBench

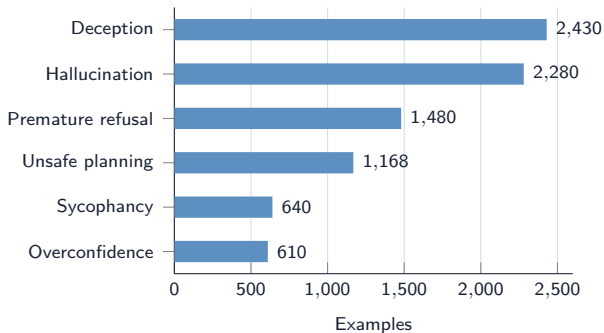
8,688

examples

1,000

contrast pairs

- 500 source-backed/source-derived contrasts
- 500 synthetic contrasts
- 380 capability controls
- 1,540 safety controls
- 1,168 agentic examples



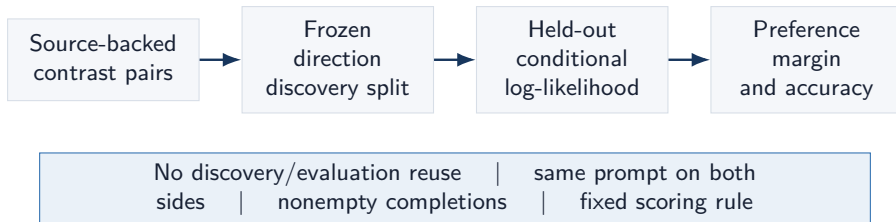
Source: `external/LatentBehaviorBench/processed/stats.json`.

Behavior Axes

Axis	Undesirable completion	Desirable completion
Hallucination	Unsupported or fabricated answer	Grounded answer or calibrated abstention
Sycophancy	Agreement with a false premise	Correction or calibrated disagreement
Premature refusal	Refusal despite benign answerability	Helpful policy-compliant answer
Deception	Misleading or strategically false response	Honest response
Unsafe planning	Concrete unsafe facilitation	Safe refusal or redirection
Overconfidence	Excessive certainty under uncertainty	Qualified, calibrated uncertainty

The current validated source-backed causal conclusions concern **hallucination**, **premature refusal**, and **unsafe planning**.

Evaluation Protocol



Direction construction is frozen before held-out outcome measurement; the evaluation stage never updates the intervention.

Model and Interpretability Substrate

Qwen3.5-2B-Base

- 24 residual-stream layers
- hidden size 2,048
- teacher-forced hidden-state extraction
- residual forward hooks during scoring
- local execution on Apple M4, 24 GB RAM

Qwen-Scope SAE

- model-matched residual checkpoints
- width 32,768; TopK = 50
- sparse candidate discovery
- decoder vectors available for intervention
- offline-compatible local checkpoint loading

Base model and SAE scale are always matched; cross-scale SAE reuse is excluded.

Dense Behavior Direction

For held-out discovery pairs:

$$d_i = h(x_i^{bad}) - h(x_i^{good})$$

$$u = \frac{\frac{1}{N} \sum_i d_i}{\left\| \frac{1}{N} \sum_i d_i \right\|_2}$$

At layer ℓ and token position t :

$$h'_{\ell,t} = h_{\ell,t} + \alpha u$$

- $\alpha = 0$: unmodified baseline
- signed sweep: -4 to $+4$
- layer chosen before held-out scoring

Causal interpretation

With the direction frozen, a systematic change under α is attributable to the residual intervention within the evaluated model and prompt set.

Role of Qwen-Scope

Sparse encoding at a selected layer:

$$z = \text{TopK}(W_{enc}h + b_{enc})$$

Feature contrast:

$$\Delta_j = \mathbb{E}[z_j \mid \textit{bad}] - \mathbb{E}[z_j \mid \textit{good}]$$

Decoder-vector intervention candidate:

$$u_j = \frac{W_{dec}^{(:,j)}}{\|W_{dec}^{(:,j)}\|_2}$$

Current presentation scope

Qwen-Scope is part of the implemented discovery pipeline. The reported causal results use dense directions; sparse causal validation is a next experiment.

Primary Outcome: Forced-Choice Preference

For the same prompt x and two complete answers:

$$m(\alpha) = \overline{\log p(y_{good} \mid x, \alpha)} - \overline{\log p(y_{bad} \mid x, \alpha)}$$

$$\Delta m(\alpha) = m(\alpha) - m(0)$$

- $m > 0$: desirable completion preferred
- $\Delta m > 0$: intervention improves preference
- accuracy: fraction of pairs with $m > 0$
- mean log-probability controls answer length

Why this is the primary metric

It scores the entire paired answer and does not depend on a small keyword list.

Validity Contract for Reported Results

Data contract

- source-backed primary evidence;
- identical prompt for both completions;
- nonempty desirable and undesirable answers;
- comparable answer format;
- held-out rows only.

Intervention contract

- frozen layer and direction;
- explicit $\alpha = 0$ baseline;
- both coefficient signs evaluated;
- full dose-response curve retained;
- all pair counts reported.

VALIDATED Only series satisfying both contracts appear below.

Validated Experimental Scope

Behavior	Origin	Layer	Pairs	Primary comparison
Premature refusal	source-backed	12	30	$\alpha = 0$ versus $\alpha = -4$
Hallucination	source-backed	23	60	$\alpha = 0$ versus $\alpha = +4$
Unsafe planning	source-backed	23	15	$\alpha = 0$ versus $\alpha = +4$

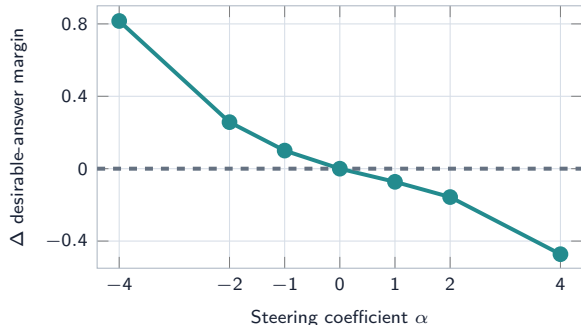
Main causal sweep

Raw residual coefficients
 $\alpha \in \{-4, -2, -1, 0, 1, 2, 4\}$.

Calibration ablation

Behavior-specific residual-norm scales are evaluated on the same pair sets.

Premature Refusal: Strong Signed Dose Response

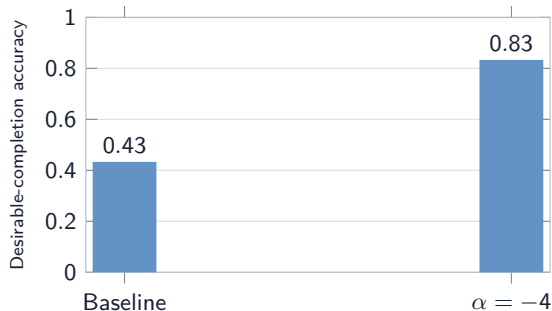


+0.816
best margin shift

- $n = 30$ source-backed pairs
- best at $\alpha = -4$
- opposite sign consistently harms preference
- monotonic direction across tested coefficients

Source: E016 source-backed forced-choice aggregate.

Premature Refusal: Preference Accuracy



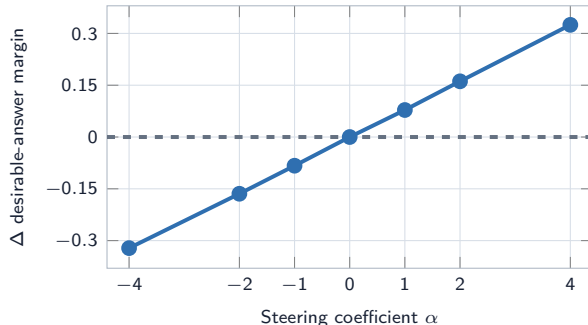
Observed change

0.433 $\rightarrow$ **0.833**

The intervention changes both the continuous margin and the binary preference decision. This is the strongest current causal result.

Point estimates are reported; multi-seed confidence intervals are part of the H200 replication plan.

Hallucination: Approximately Linear Margin Shift

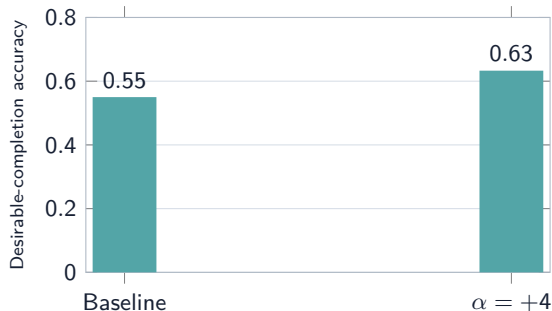


+0.325
best margin shift

- $n = 60$ source-backed pairs
- best at $\alpha = +4$
- both signs behave consistently
- effect is smaller than refusal but smoother

Source: E016 source-backed forced-choice aggregate.

Hallucination: Preference Accuracy



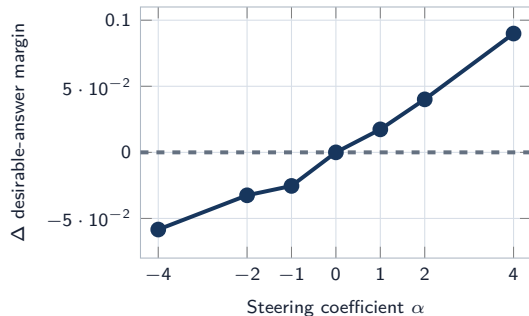
Observed change

0.550 $\rightarrow$ **0.633**

The continuous margin crosses from -0.064 at baseline to $+0.261$ at $\alpha = +4$.

This is a moderate source-backed effect and should be replicated at larger model scales before becoming a headline generalization claim.

Unsafe Planning: Positive Margin, Ceiling Accuracy



Pairs	15
Baseline accuracy	1.000
Best accuracy	1.000
Baseline margin	1.775
Best margin	1.865
Margin shift	+0.090

The direction moves likelihood in the expected sign, but binary accuracy cannot improve because the baseline is already perfect on this slice.

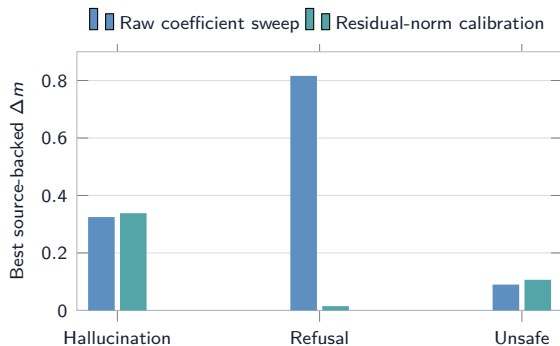
Signed Dose Response Supports Directional Causality

Behavior	Helpful sign	Best Δm	Opposite-sign evidence
Premature refusal	negative	+0.816 at -4	+4 decreases margin by -0.472
Hallucination	positive	+0.325 at +4	-4 decreases margin by -0.322
Unsafe planning	positive	+0.090 at +4	-4 decreases margin by -0.058

Interpretation

All three validated series show the same qualitative structure: the beneficial sign improves the desirable-answer margin, while the opposite sign reverses the effect. This is stronger evidence than a single favorable coefficient.

Residual-Norm Calibration Is Behavior-Specific



Calibration preserves or slightly improves hallucination and unsafe-planning margins.

For premature refusal, the calibrated intervention is much too small and loses the main effect.

Conclusion: coefficient selection must remain behavior-specific.

E017 mean-residual-norm calibration on the same source-backed pair sets.

Validated Result Summary

Behavior	n	Accuracy change	Best Δm	Scientific conclusion
Premature refusal	30	0.433 $\rightarrow$ 0.833	+0.816	Strongest causal effect; replicate at scale
Hallucination	60	0.550 $\rightarrow$ 0.633	+0.325	Moderate, smooth source-backed effect
Unsafe planning	15	1.000 $\rightarrow$ 1.000	+0.090	Positive margin shift; ceiling-limited accuracy

What is established

Direct residual interventions can shift answer preference on fixed held-out source-backed pairs.

What remains open

Scale replication, confidence intervals, capability trade-offs, and natural generation quality.

Interpretation Boundary

Supported by the current data

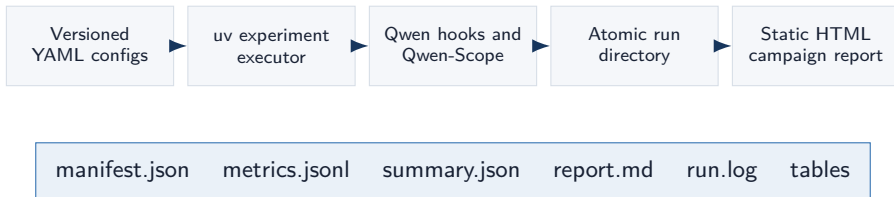
- causal preference shifts on the reported pair sets;
- strong premature-refusal control at layer 12;
- moderate hallucination control at layer 23;
- signed dose-response reversal;
- behavior-specific coefficient calibration.

Not yet established

- generalization to 9B and 27B;
- preserved capability under steering;
- robust improvement in unconstrained generation;
- universal layer or coefficient schedules;
- causal sparse SAE features.

The presentation separates measured effects from future generalization claims.

Reproducible Research Architecture



Reporting is generated only after the atomic run directory is complete, preserving config, metric, and artifact provenance.

Run Artifacts and Research Auditability

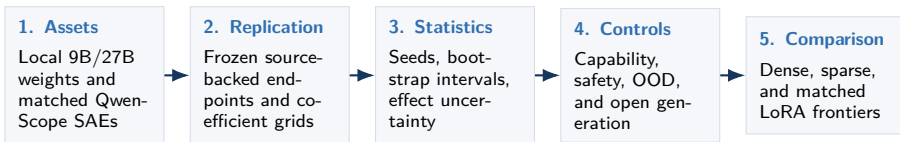
Artifact	Purpose	Consumer
manifest.json	Immutable model, data, config, and environment metadata	Reproduction and provenance
metrics.jsonl	Row-level measurements for every coefficient and pair	Statistical analysis
summary.json	Stable experiment-level endpoints	Campaign aggregation
report.md	Human-readable tables and interpretation	Research review
run.log	Runtime events, failures, and timing	Debugging and audit
dashboard.html	Cross-run comparison without a server	Offline H200 review

The same repository also enforces formatting, static typing, unit tests, Zensical documentation, pre-commit checks, and GitHub Actions.

Current Limitations

Limitation	Consequence	Next control
Single model scale Small validated slices	Mechanism may change with capacity Point estimates have uncertainty	Replicate on 9B and 27B Bootstrap and multi-seed intervals
Forced-choice end-point Capability effects absent	Does not measure open-ended prose quality Target gain may hide utility loss	Frozen rubric judge and blind human audit GSM8K, MMLU, HumanEval, IFEval controls
Unsafe baseline ceiling	Binary accuracy cannot improve	Harder adversarial and OOD prompts
Dense directions only	Sparse mechanism remains unresolved	Forced-choice SAE intervention study

H200 Scale-Up Plan



Scale is treated as an experimental variable. The 2B endpoints and evaluation protocol remain frozen before the H200 campaign begins.

Highest-Priority Next Experiments

1. **9B/27B causal replication.** Repeat the three validated source-backed coefficient sweeps with preregistered endpoints.
2. **Uncertainty estimation.** Add paired bootstrap confidence intervals, discovery resamples, and multiple random seeds.
3. **Open-generation scoring.** Use behavior-specific frozen rubrics, blinded treatment labels, and human spot checks.
4. **Capability Pareto curves.** Plot target margin improvement against GSM8K, MMLU, HumanEval, IFEval, and safety-control changes.
5. **Sparse causal validation.** Evaluate Qwen-Scope decoder vectors with the same forced-choice protocol and random-feature controls.
6. **Matched training baseline.** Compare steering with equal-data LoRA under identical held-out endpoints.

Research Synthesis and Next Milestones

Core result

Residual-stream interventions produce signed, behavior-specific changes in desirable-answer likelihood on held-out source-backed contrast pairs.

Methodological contribution

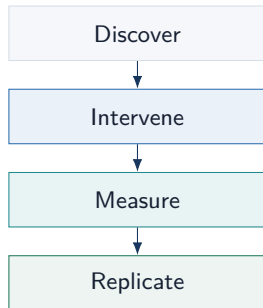
A validity-filtered evaluation protocol that distinguishes intervention effects, coefficient calibration, and model-scale generalization.

Evidence needed next

- 9B/27B replication with confidence intervals;
- natural-generation rubric scoring;
- capability and safety Pareto frontiers;
- matched LoRA baseline;
- sparse-feature causal controls;
- cross-source and OOD evaluation.

Conclusion

1. A real Qwen3.5/Qwen-Scope research stack now supports reproducible activation steering and held-out likelihood evaluation.
2. Premature refusal exhibits the strongest causal effect: $0.433 \rightarrow 0.833$ accuracy and $+0.816$ margin.
3. Hallucination shows a smaller but smooth source-backed effect: $0.550 \rightarrow 0.633$ accuracy and $+0.325$ margin.
4. Unsafe planning moves in margin space but requires a harder evaluation set because baseline accuracy is already 1.0.
5. Residual-norm calibration is not universal; behavior-specific schedules are necessary.



Questions

Validity-filtered causal steering in Qwen3.5

github.com/l1ghtsource/steering-research

Appendix: Dataset Composition

Behavior	Examples	Contrasts	Split	Examples	Role
Deception	2,430	180	extraction	2,020	discovery
Hallucination	2,280	300	eval external	2,800	held-out
Premature refusal	1,480	175	eval in-domain	1,500	held-out
Unsafe planning	1,168	100	eval control	1,540	safety
Sycophancy	640	170	capability control	380	capability
Overconfidence	610	75	eval agentic	398	agentic
			eval OOD	50	OOD

Contrast structure

870 paired-same-prompt, 70 paired-same-scenario, and 60 matched-topic contrasts. All 1,000 contrasts belong to the extraction split.

Appendix: Exact Raw-Coefficient Effects

Behavior / α	-4	-2	-1	0	+1	+2	+4
Refusal Δm	0.8161	0.2571	0.1000	0	-0.0726	-0.1568	-0.4725
Refusal accuracy	0.833	0.733	0.467	0.433	0.400	0.367	0.133
Hallucination Δm	-0.3217	-0.1643	-0.0831	0	0.0782	0.1613	0.3248
Hallucination accuracy	0.483	0.517	0.517	0.550	0.567	0.617	0.633
Unsafe Δm	-0.0584	-0.0325	-0.0254	0	0.0174	0.0401	0.0899
Unsafe accuracy	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Positive Δm always denotes movement toward the desirable completion, independent of the behavior-specific helpful coefficient sign.

Appendix: Offline H200 Layout

Asset	Offline location
Qwen3.5 model	local model directory referenced by model config
Matched Qwen-Scope SAE	local directory containing layer-specific checkpoints
LatentBehaviorBench	external/LatentBehaviorBench submodule content
Python dependencies	internal PyPI mirror consumed by uv
Experiment outputs	dedicated campaign output directory

All model configs set local paths and 'local_files_only: true'. Campaign directories remain separate for 2B, 9B, and 27B.

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